

FUAD AHMED (TUBA, FUAD AHMED)

STATEMENT OF PURPOSE & MOTIVATION LETTER

Target University: Politecnico di Milano — School of Industrial and Information Engineering

Degree Program: Laurea Magistrale (MSc) in Computer Science and Engineering (Ingegneria Informatica)

Campus: Milano Leonardo • **Applicant:** fuadahmedt@gmail.com • +251 925 278 350 • Addis Ababa, Ethiopia

To the Academic Admissions Committee of Politecnico di Milano,

I am writing to formally submit my application for the **Laurea Magistrale in Computer Science and Engineering** at **Politecnico di Milano** for the 2027/2028 academic year. As a software systems engineer with four years of hands-on experience building high-concurrency backend services and distributed data architectures, my goal is to deepen my theoretical foundations and rigorous engineering methodologies under the mentorship of Politecnico di Milano's internationally distinguished faculty. I am particularly drawn to the program's specialized curricula in Distributed Systems, High-Performance Computing, and Cloud Infrastructure.

1. ACADEMIC TRAJECTORY & COMPUTATIONAL FOUNDATIONS

I earned my Bachelor of Science in Computer Science from St. Mary's University in Addis Ababa, graduating with a Cumulative GPA of 3.20/4.00 within a rigorous four-year curriculum conducted entirely in English. My undergraduate coursework provided me with solid mathematical competence across Discrete Mathematics, Linear Algebra, Probability & Statistics, and Calculus, alongside core systems disciplines including Operating Systems, Computer Architecture, Distributed Systems, and Advanced Algorithms.

For my senior capstone project, I led the architecture and empirical evaluation of a fault-tolerant, high-concurrency distributed data synchronization protocol. Operating across networked Linux nodes, my protocol utilized delta-encoded state serialization and adaptive consensus verification to maintain data consistency while minimizing thread contention and network overhead. This capstone solidified my ambition to specialize in large-scale distributed computing systems.

2. PROFESSIONAL ENGINEERING EXPERIENCE & MOTIVATION

Over the past four years, I have worked as a Lead Backend and Distributed Systems Software Engineer, developing enterprise-grade microservices and data pipelines using Python (FastAPI/AsyncIO), Go, and Docker. In production environments, I designed distributed Redis caching topologies that reduced database read latency by 48% and implemented asynchronous message-driven architectures using Kafka and RabbitMQ to guarantee system fault-tolerance during sudden load spikes.

While industry practice has honed my practical capabilities in Linux kernel tuning, container orchestration, and concurrent network programming, it has also revealed the necessity of deep theoretical mastery. Modern distributed computing confronts fundamental limits in consensus scalability, tail-latency bounds, and memory consistency models. Politecnico di Milano's rigorous balance between formal computer science theory and rigorous engineering implementation provides the ideal crucible to master these challenges.

3. WHY POLITECNICO DI MILANO & DIPARTIMENTO DI ELETTRONICA, INFORMAZIONE E BIOINGEGNERIA (DEIB)

Politecnico di Milano stands as Italy's preeminent technological university and ranks among the top engineering institutions worldwide. The legacy of Luigi Dadda and the pioneering work conducted within the Dipartimento di Elettronica, Informazione e Bioingegneria (DEIB) represent the gold standard in computing research.

I am eager to engage with the advanced course offerings at the Milano Leonardo campus, specifically *Distributed Systems*, *Advanced Computer Architectures*, *Cloud Computing & Distributed Middleware*, and *High-Performance*

Computing. Furthermore, I am inspired by the cutting-edge research conducted at laboratories such as the NECSTLab (Novel, Emerging Computing System Technologies Laboratory), where heterogeneous computing architectures, memory-centric systems, and resilient distributed platforms are developed. Collaborating with faculty who bridge foundational systems research with European industrial innovation is the exact environment where I can thrive.

4. CAREER VISION & COMMITMENT

Following completion of my Master of Science degree at Politecnico di Milano, I plan to pursue advanced research and development roles as a Distributed Systems Architect within European technology enterprises or research laboratories, designing resilient, low-latency cloud and edge infrastructures. In the long term, I intend to leverage this elite European training to contribute to the modernization of distributed digital infrastructure in emerging economies.

I bring to Politecnico di Milano a proven track record of engineering perseverance, academic discipline, and a deep appreciation for the Italian technological heritage. I am fully prepared to meet the demanding standards of the MSc in Computer Science and Engineering.

Sincerely and respectfully,

Fuad Ahmed (TUBA, FUAD AHMED)

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